

Maíme dokázať následujúce tvrzenie:

$$\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2),$$

m. d/e vieme že platí $\operatorname{Re}(x) = \frac{x + \bar{x}}{2}$

$$\forall x \in \mathbb{C} : \operatorname{Re}(x) = a \quad \text{dikaz:} \quad \frac{x + \bar{x}}{2} = \frac{a+bi + a-bi}{2} = \underline{\underline{a}}$$

$\downarrow$
 $x = a+bi$

a budeme tak chcieť prokázat že se pravá strana tvrzení bude rovnat: $\frac{x + \bar{x}}{2}$ kde za x máme dosadiť $\langle \mathbf{u}, \mathbf{v} \rangle$

$$\text{takže: } \frac{\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle}}{2}$$

počítajme

$$\begin{aligned} \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) &= \frac{1}{4}(\langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle) \\ &= \frac{1}{4}[\langle \mathbf{u}, \mathbf{u} + \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - (\langle \mathbf{u}, \mathbf{u} - \mathbf{v} \rangle + \langle -\mathbf{v}, \mathbf{u} - \mathbf{v} \rangle)] \\ &= \frac{1}{4}[\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - (\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, -\mathbf{v} \rangle + \langle -\mathbf{v}, \mathbf{u} \rangle + \langle -\mathbf{v}, -\mathbf{v} \rangle)] \\ &= \frac{1}{4}(\cancel{\langle \mathbf{u}, \mathbf{u} \rangle} + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \cancel{\langle \mathbf{v}, \mathbf{v} \rangle} - \cancel{\langle \mathbf{u}, \mathbf{u} \rangle} + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle - \cancel{\langle \mathbf{v}, \mathbf{v} \rangle}) \\ &= \frac{1}{4}(2\langle \mathbf{u}, \mathbf{v} \rangle + 2\langle \mathbf{v}, \mathbf{u} \rangle) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle}}{2} \Rightarrow = \operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

$\downarrow$
 $\langle \mathbf{v}, \mathbf{u} \rangle = \overline{\langle \mathbf{u}, \mathbf{v} \rangle}$

což je náš cíl
dikaz hotov

Maíme dokázat následující tvrzení:

$$\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2),$$

my ale víme že platí

$$\operatorname{Im}(x) = \frac{x - \bar{x}}{2i}$$

$$\forall x \in \mathbb{C}: x = a + bi: \operatorname{Im}(x) = b \quad ; \text{ díky } \frac{x - \bar{x}}{2i} = \frac{a + bi - a + bi}{2i} = \underline{\underline{b}}$$

a budeme tak chtít prokázat že se pravá strana tvrzení bude rovnat: $\frac{x - \bar{x}}{2i}$ kde za x můžeme dosadit $\langle \mathbf{u}, \mathbf{v} \rangle$

$$\text{takže: } \frac{\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle}}{2i}$$

pocítejme

$$\begin{aligned} \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2) &= \frac{1}{4}(\langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle) \\ &= \frac{1}{4}[\langle \mathbf{u}, \mathbf{u} - i\mathbf{v} \rangle + \langle -i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - (\langle \mathbf{u}, \mathbf{u} + i\mathbf{v} \rangle + \langle i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle)] = \frac{1}{4}[\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, -i\mathbf{v} \rangle + \\ &\quad \langle -i\mathbf{v}, \mathbf{u} \rangle + \langle -i\mathbf{v}, -i\mathbf{v} \rangle - (\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, i\mathbf{v} \rangle + \langle i\mathbf{v}, \mathbf{u} \rangle + \langle i\mathbf{v}, i\mathbf{v} \rangle)] \\ &= \frac{1}{4}(\cancel{\langle \mathbf{u}, \mathbf{u} \rangle} - i\langle \mathbf{u}, \mathbf{v} \rangle + i\langle \mathbf{v}, \mathbf{u} \rangle + \cancel{\langle \mathbf{v}, \mathbf{v} \rangle} - \cancel{\langle \mathbf{u}, \mathbf{u} \rangle} - i\langle \mathbf{u}, \mathbf{v} \rangle + i\langle \mathbf{v}, \mathbf{u} \rangle - \cancel{\langle \mathbf{v}, \mathbf{v} \rangle}) \\ &= \frac{1}{4}(-2i\langle \mathbf{u}, \mathbf{v} \rangle + 2i\langle \mathbf{v}, \mathbf{u} \rangle) = \frac{-i}{2}(\langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle) = \\ &= \frac{-i}{2}(\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) \cdot \frac{-i}{-i} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle}}{2i} \Rightarrow = \operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

rozšíření

dokázáno!

využít operace:

pozorování 8.15

definice 8.14

skripta

definice 8.9